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Materials for Energy (ChE 494)

Quiz 1: Science of Materials

February 11, 2020 **40 Minutes** **Total: 100**

1. How to construct a crystal structure? Briefly define lattice and basis.

Lattice is the form that the basis (atoms) takes to Make a crystal structure.

2. No property of solids varies as widely as their ability to conduct electric current.

(A) True (B) False

3. At absolute zero, perfect crystals of most semiconductors will be an insulator.

(A) True (B) False

4. What are the 7 crystal systems those constitute 14 different lattice types in 3D?

5. Graphene has an exceptionally well-formed orthorhombic lattice structure.

(A) True (B) False

6. How many lattices are in a cubic system? Write the names and draw those lattices?

7. Fullerene-Graphite-Carbon Nanotubes-Graphene (Arrange these materials with the following dimensions in the same order).

(A) 3D-2D-0D-1D (B) 0D-3D-1D-2D (C) 2D-1D-0D-3D (D) 0D-2D-1D-3D

8. How many "surfaces" are in a monolayer of 2D nanomaterial (For Example: Boron Nitride)?

(A) 1 (B) 2 (C) 3 (D) 6

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9. 0D Confinement-2D Confinement-1D Confinement-3D Confinement (Confinement vs. Materials).

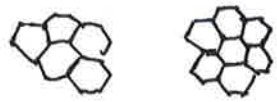
   (A) 3D-1D-2D-0D (B) 0D-1D-3D-2D (C) 2D-1D-3D-0D (D) 3D-1D-2D-0D

10. How many lattice points per unit cell of a face-centered cubic lattice?

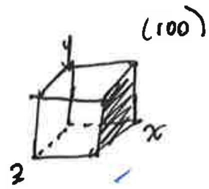
    (A) 1 (B) 2 (C) 4 (D) 6

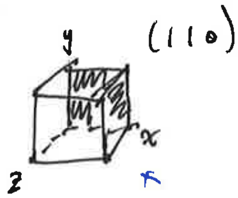
11. The number of atoms in each hexagonal unit cell of a graphene lattice is 6.

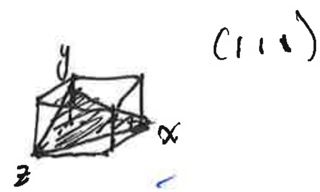
    (A) True (B) False



12. Draw the (100), (110), and (111) crystal planes?







13. In MoS_2, the atoms are held together by strong intra-layer covalent bonds. However, the layer interactions (inter-layer bonds) are weak and are held loosely by van der Waals forces.

    (A) True (B) False

14. Monolayer Boron Nitride, Monolayer MoS_2, Monolayer Graphene (Match these 2D materials with the following energy bandgaps in the same order).

    (A) 1.8 eV, 0.5 eV, 6 eV (B) 0 eV, 3 eV, 1.8 eV (C) 10 eV, 0 eV, 1.8 eV (D) 6 eV, 1.8 eV, 0 eV

15. Insulator + 0.34 nm Thickness + sp^2 Lattice (Match these properties with one of the following 2D materials).

    (A) MoS_2 (B) Boron Nitride (C) Graphene (D) WS_2

16. Which polytype of MoS_2 crystal structure represents: rhombohedral symmetry, three layers per repeat unit, and trigonal prismatic coordination?

    (A) 1T (B) 3R (C) 2H (D) None

17. Boron nitride (BN) is a synthetic binary compound located in III and V group elements in the Periodic Table.

    (A) True (B) False

18. In graphene, the sigma (σ) bonds are delocalized electrons and pi (π) bonds are localized electrons.

    (A) True (B) False

19. In an indirect bandgap semiconductor, an electron can shift from the highest-energy state in the valence band (VB) to the lowest-energy state in the conduction band (CB) without a change in the crystal momentum.

(A) True (B) False

20. A polycrystalline solid is made up of an aggregate of many small single crystals.

(A) True (B) False

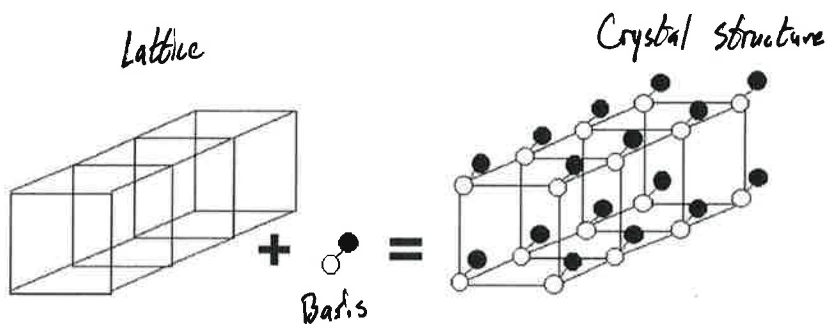
21. Which 2D material can be used for deep ultraviolet (UV ≈ 6 eV) emission or detection?

(A) Graphene (B) Boron Nitride (C) MoS_2 (D) WS_2

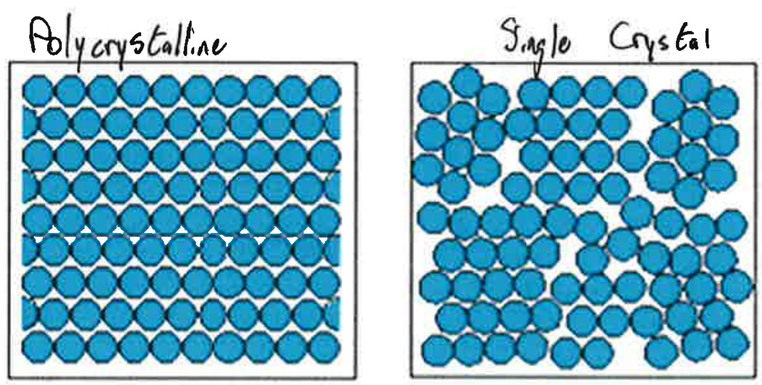
22. Graphene is isostructural and isoelectronic to………………………………………………………………

(A) MoS_2 (B) h-BN (C) Carbon Nanotube (D) Fullerene (C_{60})

23. Designate lattice, basis, and crystal structure in the following figure.



24. Differentiate between single crystal and polycrystalline solids from the following figure.



25. A crystalline solid is the solid form of a substance in which the atoms or molecules are arranged in a random pattern.

(B) True (B) False

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Optional Question(s):

26. What are the rules to index the crystal planes? Write down the rules?

27. Explain the tight binding methods for energy bands considering two Hydrogen atoms?

28. Characteristics of cubic lattices.

	Simple Cubic	Body-Centered Cubic	Face-Centered Cubic
Unit Cell Volume	Vol = a^3 , 2	Vol = a^3 , 3	vol = a^3 , 4
Lattice Points Per Unit Cell	8	9	14
Number of Nearest Neighbors	3	4	